One Picture, Three Ways

Name: _____ Date: _____

Every `Pict` has methods that mimic corresponding constructors: `a.beside(b)` is `new Beside(a, b)`, `a.above(b)` is `new Above(a, b)`, and `a.overlay(b)` is `new Overlay(a, b)`, with `a` in front.

nested constructors

```
Pict scene = new Beside(
    new Above(top, trunk),
    sun);
```

named steps

```
Pict tree = new Above(top,
trunk);
Pict scene = new Beside(tree,
sun);
```

chained methods

```
Pict scene =
top.above(trunk).beside(sun);
```

All three definitions start from these pieces:

```
Pict trunk = new Rect(10, 20, "brown");
Pict top = new Triangle(30, 40, "green");
Pict sun = new Circle(8, "yellow");
Pict disc = new Circle(30, "red");        // overlaying disc on field gives us 🇧🇩
Pict field = new Rect(150, 100, "green");
```

Three Styles

Q1. Write each picture in the two styles it is not shown in.

nested	named steps	chained
`new Above(new Above(sun, top), trunk)`		
		`disc.overlay(field)`
	`Pict bluewhite = new Beside(blue, white);` `Pict flag = new Beside(bluewhite, red);`	

Q2. Java prints records as `Constructor[field=val,...]`. For the three `scene` definitions at the top of the sheet, write what it prints for `scene`.

Reading a Chain

A chain is read left to right, and each step uses everything to its left as the receiver. Parentheses change which picture is the receiver.

Q3. Draw the records each expression builds and say in words what the picture looks like.

```
top.above(trunk).beside(sun)
```

```
top.above(trunk.beside(sun))
```

Same picture, same printout?

Q4. For each pair, do the two print the same line? Do they draw the same picture?

pair	same printout?	same drawing?
`top.above(trunk)` and `new Above(top, trunk)`		
`new Beside(sun, top)` and `new Beside(top, sun)`		
`new Below(trunk, top)` and `new Above(top, trunk)`		
`sun.beside(top).beside(trunk)` and `sun.beside(top.beside(trunk))`		

Q5. Two rows draw the same picture but print differently. If a classmate showed you only their picture, name one thing the printout would tell you that the picture does not.

Which would you write?

Here is a scene: a house (a triangle roof above a square wall), with a tree (from the top of the sheet) on its left and the sun above everything.

Q6. Write it out. Which style did you pick? Did you mix styles at all?